

Perbaikan & Penyelarasan Learning Outcomes

Program Studi Data Science -- Diselaraskan dengan Kebutuhan Industri & Career Development Center (CDC)

8 Semester + 2 Short Semester

152 SKS Total

3 Concentration Tracks

MBKM Semester 6 & 7

CDC-Aligned Tools

Ringkasan Perbaikan

Perubahan Strategis yang Dilakukan

- Merestrukturisasi learning outcomes agar **berbasis kompetensi terukur** (menggunakan kata kerja Bloom's Taxonomy: Apply, Implement, Design, Evaluate).
- Memperkuat mata kuliah semester awal dengan **proyek praktik langsung** agar mahasiswa memiliki portofolio dan siap kerja.
- Menambahkan **jalur konsentrasi** di Semester 5 untuk spesialisasi: Data Engineering, AI/ML, Business Intelligence.
- Mengintegrasikan **program MBKM** di Semester 6 dan 7 untuk pengalaman industri langsung.
- Menambahkan **output terukur per semester**: mini-project, portofolio, capstone project, dan publikasi ilmiah.
- Menyelaraskan kurikulum dengan **kebutuhan CDC Fakultas IT** sehingga lulusan siap memasuki pasar kerja.
- Menambahkan **Learning Tools** yang relevan dengan industri pada setiap mata kuliah, diselaraskan dengan kebutuhan CDC.

Kebutuhan CDC Fakultas IT

Posisi Kerja & Tools yang Dibutuhkan

1 Software Engineering JavaScript, Python, Java, React, Next.js, Laravel, Node.js, GitHub, Docker	2 Web & CMS Development WordPress, Elementor, HTML, CSS, JavaScript	3 DevOps & Infrastructure Docker, Kubernetes, Jenkins, GitHub Actions, AWS, GCP, Azure, Terraform
4 Data & Analytics SQL, Excel, Python (Pandas), Power BI, Tableau	5 QA & Testing Test case tools, Selenium	6 System & Business Analyst Draw.io, Lucidchart, Notion, Confluence
7 UI/UX & Product Design Figma, Adobe XD, Maze	8 Product Management Notion, Jira, Google Docs	9 IT Security & Cybersecurity Kali Linux, Nmap, Wireshark
10 IoT Arduino, ESP32, MQTT	11 Game Development Unity, Unreal Engine	12 IT Support & Compliance OS knowledge, Basic networking tools
13 Solution Engineer Postman (API), CRM/ERP tools		

SEMESTER 120 SKS • Fondasi Data Science & Kompetensi Dasar

NO	MATA KULIAH	LEARNING OUTCOMES	LEARNING TOOLS
1	Foundations of Data Science 3 SKS	1. Apply fundamental concepts of data science to real-world problems 2. Implement basic statistical methods for data analysis 3. Develop data processing pipelines 4. Interpret results of data analysis for decision support 5. Evaluate data sources for quality and relevance	PythonJupyter Notebook Google ColabPandasNumPy
2	Design Thinking 2 SKS	1. Apply design thinking methodology to identify and solve problems 2. Demonstrate empathy in understanding user needs and requirements 3. Generate innovative solutions through ideation techniques 4. Develop and test prototypes iteratively 5. Evaluate solutions based on user feedback and technical feasibility	FigmaMiroDraw.ioNotion
3	Discrete Mathematics 3 SKS	1. Apply mathematical logic and proof techniques in problem-solving 2. Analyze combinatorial problems using appropriate counting techniques 3. Understand set theory and its applications in computing 4. Model problems using graph theory and relations 5. Apply discrete mathematical concepts to computational problems	PythonMATLABLaTeX
4	Human-Computer Interaction 3 SKS	1. Apply principles of user-centered design in interface development 2. Analyze user requirements and translate them into interface specifications 3. Design interfaces that optimize usability and user experience 4. Evaluate interfaces using appropriate testing methodologies 5. Apply accessibility standards in interface design	FigmaAdobe XD Maze InVision
5	Algorithms 3 SKS	1. Analyze algorithmic problems and develop efficient solutions 2. Implement classic algorithms for sorting, searching, and graph traversal 3. Apply algorithm design techniques to solve computational problems 4. Analyze the time and space complexity of algorithms 5. Evaluate and optimize algorithmic solutions	PythonC++LeetCode GitHub
6	Mathematical and Statistical Foundations 2 SKS	1. Apply mathematical concepts essential for data science 2. Implement statistical tests for hypothesis testing 3. Apply probability theory in modeling uncertainty 4. Interpret statistical results in context 5. Develop statistical models for data analysis	PythonRExcelSPSS
7	Networks 2 SKS	1. Explain fundamental network concepts and architectures 2. Apply network protocols for data science applications 3. Analyze network data and traffic patterns 4. Implement basic network algorithms and models 5. Evaluate network performance and reliability	WiresharkCisco Packet Tracer Nmap
8	Differential Calculus 2 SKS	1. Apply differentiation techniques to mathematical functions 2. Analyze rates of change and optimization problems 3. Implement numerical differentiation algorithms 4. Apply calculus concepts to data science problems 5. Interpret derivatives in the context of data analysis	PythonMATLAB Wolfram AlphaGeoGebra

SEMESTER 2

20 SKS • Pemrograman Lanjut, Database & Data Wrangling

NO	MATA KULIAH	LEARNING OUTCOMES	LEARNING TOOLS
1	Object-Oriented Programming 3 SKS	1. Apply object-oriented principles (encapsulation, inheritance, polymorphism) 2. Design programs using class hierarchies and relationships 3. Implement object-oriented solutions to complex problems 4. Apply design patterns in software development 5. Evaluate and refactor object-oriented code for improved maintainability	Python Java GitHub/GitLab VS Code UML Tools
2	Database Systems 3 SKS	1. Design database schemas using entity-relationship modeling 2. Implement relational and non-relational database solutions 3. Apply SQL and query optimization techniques 4. Develop data access layers for applications 5. Evaluate database performance and scalability	MySQL PostgreSQL MongoDB SQL DBeaver
3	Data Structures 3 SKS	1. Implement fundamental data structures for efficient data organization 2. Apply appropriate data structures for different problem domains 3. Analyze the time and space complexity of data structure operations 4. Implement custom data structures for specific requirements 5. Evaluate data structure selection impact on algorithm performance	Python Java C++ GitHub
4	Web Client Development 3 SKS	1. Develop responsive web interfaces using HTML, CSS, and JavaScript 2. Apply front-end frameworks to build interactive web applications 3. Implement client-side validation and error handling 4. Integrate web interfaces with backend services via APIs 5. Optimize web applications for performance and accessibility	HTML CSS JavaScript React Bootstrap VS Code
5	Statistical Thinking 3 SKS	1. Apply statistical inference techniques to draw conclusions from data 2. Design experiments and observational studies 3. Implement resampling methods for uncertainty estimation 4. Interpret confidence intervals and hypothesis tests 5. Evaluate statistical models for goodness of fit	Python R SPSS Pandas Matplotlib
6	Communication Protocols 3 SKS	1. Apply communication protocol principles for data transmission 2. Implement data encoding and decoding techniques 3. Analyze protocol performance and reliability 4. Design communication protocols for specific applications 5. Evaluate protocol security and efficiency	Wireshark Postman cURL MQTT
7	Data Wrangling 2 SKS	1. Clean and preprocess raw data for analysis 2. Transform data into appropriate formats for different analyses 3. Handle missing data using appropriate techniques 4. Integrate data from multiple sources 5. Automate data preparation workflows	Python Pandas NumPy OpenRefine Excel

SHORT SEMESTER 1

9 SKS • Independent Study / Remedial / Elective

NO	MATA KULIAH	KETERANGAN	LEARNING TOOLS
1	Independent Study Project 9 SKS	Proyek mandiri berbasis data science	Sesuai Topik GitHub Python
2	Remedial Course 9 SKS	Mata kuliah remedial (bila diperlukan)	Sesuai MK
3	Elective Course 9 SKS	Mata kuliah pilihan sesuai minat	Sesuai MK

SEMESTER 3

18 SKS • Full-Stack Development, Visualisasi & Metode Numerik

NO	MATA KULIAH	LEARNING OUTCOMES	LEARNING TOOLS
1	Web Application Development 3 SKS	1. Design and implement full-stack web applications using modern frameworks and technologies 2. Apply MVC architecture principles and design patterns in web development 3. Implement authentication, authorization, and security measures in web applications 4. Develop RESTful web services and APIs for client-server communication 5. Deploy and maintain web applications in production environments	Node.js React Next.js Laravel Express.js GitHub
2	Advanced Database Systems 3 SKS	1. Design and implement advanced database solutions for complex data management requirements 2. Apply optimization techniques for database performance in high-volume environments 3. Implement transaction management and concurrency control mechanisms 4. Design and develop NoSQL database solutions for specific use cases 5. Evaluate database architectures for scalability, reliability, and security	PostgreSQL MongoDB Redis Cassandra DBeaver
3	Stochastic Modeling 2 SKS	1. Apply probability theory to model uncertainty in real-world systems 2. Develop stochastic processes to represent dynamic random phenomena 3. Implement Markov chains and queuing theory models for system analysis 4. Analyze time series data using appropriate stochastic models 5. Evaluate model accuracy and make predictions with confidence intervals	Python R SciPy SimPy
4	Numerical Methods 2 SKS	1. Implement numerical solutions for mathematical problems using computational techniques 2. Apply numerical methods for solving differential equations and linear systems 3. Analyze algorithm stability, convergence, and error propagation 4. Develop efficient implementations of numerical algorithms 5. Evaluate and select appropriate numerical methods for specific problem domains	Python MATLAB NumPy SciPy
5	Advanced Computational Mathematics 2 SKS	1. Apply advanced mathematical techniques to solve computational problems 2. Implement algorithms for numerical optimization and approximation 3. Analyze mathematical models using computational approaches 4. Develop solutions for complex mathematical problems using appropriate software tools 5. Evaluate computational methods for efficiency, accuracy, and applicability	Python MATLAB Wolfram Alpha LaTeX
6	Data Visualization 3 SKS	1. Design effective visual representations of complex data sets 2. Implement interactive data visualizations using modern visualization libraries and tools 3. Apply perceptual and cognitive principles to optimize information communication 4. Develop dashboards and visual analytics solutions for different audiences 5. Evaluate visualizations for effectiveness, accuracy, and ethical considerations	Tableau Power BI D3.js Matplotlib Seaborn Plotly
7	Optimization Methods 3 SKS	1. Formulate optimization problems from real-world scenarios 2. Apply linear, nonlinear, and integer programming techniques to solve optimization problems 3. Implement metaheuristic algorithms for complex optimization challenges 4. Analyze sensitivity and robustness of optimization solutions 5. Evaluate optimization methods for computational efficiency and solution quality	Python SciPy PuLP MATLAB Gurobi

SEMESTER 4

18 SKS • Text Mining, Big Data & Advanced Analytics

NO	MATA KULIAH	LEARNING OUTCOMES	LEARNING TOOLS
1	Text Mining 3 SKS	1. Apply natural language processing techniques to extract information from unstructured text 2. Implement text preprocessing, tokenization, and normalization methods 3. Develop text classification and clustering models for document analysis 4. Apply sentiment analysis and topic modeling algorithms to text corpora 5. Evaluate text mining solutions for accuracy, scalability, and business relevance	PythonNLTKspaCy Hugging FaceScikit-learn
2	Data Warehousing and Mining 3 SKS	1. Design data warehouse architectures to support business intelligence requirements 2. Implement ETL processes for data integration from heterogeneous sources 3. Apply dimensional modeling techniques for analytical data representation 4. Develop data mining algorithms for pattern discovery and prediction 5. Evaluate data warehousing and mining solutions for performance and business value	SQLPentahoApache Spark Power BITableau
3	Technical / Professional Writing 2 SKS	1. Create clear and concise technical documentation for diverse audiences 2. Apply proper structure and organization in technical and business documents 3. Develop data-driven reports that effectively communicate analytical findings 4. Implement visualizations and formatting to enhance document readability 5. Evaluate technical documentation for clarity, accuracy, and audience appropriateness	LaTeXNotionConfluence Google DocsMarkdown
4	Mobile Computing 3 SKS	1. Design mobile applications that incorporate data science capabilities 2. Implement data collection and processing for mobile sensing applications 3. Apply offline data analytics techniques for mobile environments 4. Develop synchronization strategies for distributed mobile data 5. Evaluate mobile applications for performance, battery efficiency, and user experience	FlutterReact Native Android StudioFirebase
5	Big Data Analytics 2 SKS	1. Apply distributed computing frameworks for large-scale data processing 2. Implement parallel algorithms for big data analysis 3. Develop data pipelines for processing structured and unstructured big data 4. Apply real-time analytics techniques for streaming data 5. Evaluate big data solutions for scalability, fault tolerance, and processing efficiency	Apache SparkHadoopKafka PythonDatabricks
6	Advanced Methods for Data Analytics 3 SKS	1. Apply advanced statistical methods for complex data analysis problems 2. Implement ensemble and deep learning techniques for predictive modeling 3. Develop feature engineering strategies to improve model performance 4. Apply dimensionality reduction and regularization methods for high-dimensional data 5. Evaluate advanced analytics methods for accuracy, interpretability, and computational efficiency	PythonScikit-learn TensorFlowXGBoostMLflow
7	Data Privacy and Security 2 SKS	1. Apply privacy-preserving data mining techniques in analytics workflows 2. Implement data anonymization and pseudonymization methods 3. Develop data governance frameworks for privacy compliance 4. Apply security measures for protecting sensitive data throughout its lifecycle 5. Evaluate privacy and security solutions against regulatory requirements and ethical standards	Kali LinuxWiresharkPython OWASP ZAPNmap

SHORT SEMESTER 29 SKS • Independent Study / Remedial / Elective

NO	MATA KULIAH	KETERANGAN	LEARNING TOOLS
1	Independent Study Project 9 SKS	Proyek mandiri spesialisasi data science	Sesuai TopikGitHubPython
2	Remedial Course 9 SKS	Mata kuliah remedial (bila diperlukan)	Sesuai MK
3	Elective Course 9 SKS	Mata kuliah pilihan sesuai minat	Sesuai MK

SEMESTER 518 SKS • Concentration Tracks (pilih salah satu) + Professional Ethics

Concentration I: Data Engineering and Big Data Analytics			
NO	MATA KULIAH	LEARNING OUTCOMES	LEARNING TOOLS
1	Big Data Infrastructure 2 SKS	1. Design scalable infrastructure architectures for big data processing 2. Implement distributed storage systems for large-scale data management 3. Apply resource allocation and scheduling techniques in big data clusters 4. Configure big data platforms for high availability and fault tolerance 5. Evaluate infrastructure solutions for performance, scalability, and cost-effectiveness	HadoopApache SparkHDFS YARNMesos
2	Distributed Systems 3 SKS	1. Design distributed system architectures for data-intensive applications 2. Implement distributed algorithms for coordination and consensus 3. Apply fault tolerance and recovery mechanisms in distributed environments 4. Develop distributed data processing applications using modern frameworks 5. Evaluate distributed systems for consistency, availability, and partition tolerance	Apache KafkaZooKeeper gRPCDockerKubernetes
3	Data Pipeline Development 3 SKS	1. Design end-to-end data pipelines for diverse data sources and targets 2. Implement batch and stream processing components in data workflows 3. Apply data quality and validation techniques throughout the pipeline 4. Develop monitoring and error handling mechanisms for pipeline reliability 5. Evaluate data pipelines for throughput, latency, and fault resilience	Apache AirflowLuigiPrefect dbtPython
4	Cloud Computing 2 SKS	1. Deploy big data processing workloads on cloud platforms 2. Implement cloud-native storage solutions for data engineering 3. Apply auto-scaling and resource optimization in cloud environments 4. Develop infrastructure-as-code for reproducible cloud deployments 5. Evaluate cloud service options for cost, performance, and compliance requirements	AWSGCPAzureTerraform Docker
5	ETL Processes 3 SKS	1. Design robust ETL workflows for complex data integration scenarios 2. Implement transformation logic for data cleansing and standardization 3. Apply incremental loading and change data capture techniques 4. Develop metadata management and lineage tracking in ETL processes 5. Evaluate ETL solutions for reliability, maintainability, and performance	Apache NiFITalend InformaticaPythonSQL
6	Data Lakes 3 SKS	1. Design data lake architectures that support diverse analytical workloads 2. Implement data governance and catalog systems for data discovery 3. Apply schema-on-read approaches for flexible data analysis 4. Develop data lake processing patterns for batch and interactive queries 5. Evaluate data lake implementations for scalability, security, and analytical capabilities	AWS S3Delta Lake Apache IcebergHivePresto

Concentration II: Artificial Intelligence (AI) and Machine Learning (ML) Development

NO	MATA KULIAH	LEARNING OUTCOMES	LEARNING TOOLS
1	Deep Learning 3 SKS	1. Design and implement neural network architectures 2. Apply deep learning algorithms to solve complex problems 3. Train and optimize deep learning models 4. Evaluate deep learning models for performance and generalization 5. Implement transfer learning techniques for resource efficiency	TensorFlow PyTorch Keras CUDA Google Colab
2	Natural Language Processing (NLP) 3 SKS	1. Apply text preprocessing techniques for NLP tasks 2. Implement language models for text generation and analysis 3. Design and develop sentiment analysis applications 4. Apply named entity recognition and information extraction 5. Evaluate NLP models for accuracy and generalization	Hugging Face spaCy NLTK Transformers GPT API
3	Computer Vision 3 SKS	1. Apply image processing techniques for computer vision tasks 2. Implement object detection and recognition algorithms 3. Design systems for image classification and segmentation 4. Apply deep learning models for visual data analysis 5. Evaluate computer vision systems for performance and accuracy	OpenCV YOLO TensorFlow PyTorch Pillow
4	Reinforcement Learning 3 SKS	1. Apply reinforcement learning principles to sequential decision problems 2. Implement value-based and policy-based learning algorithms 3. Design reward functions for specific application domains 4. Develop multi-agent reinforcement learning systems 5. Evaluate reinforcement learning models for performance and stability	OpenAI Gym Stable Baselines3 PyTorch Ray RLlib
5	AI-Planning and Search Strategies 2 SKS	1. Apply search algorithms to solve planning problems 2. Implement heuristic search techniques for efficiency 3. Design planning representations for complex domains 4. Develop constraint satisfaction problem solvers 5. Evaluate planning algorithms for completeness and optimality	Python PDDL A* Search NetworkX OR-Tools
6	Model Deployment 2 SKS	1. Deploy machine learning models to production environments 2. Implement model serving architectures and APIs 3. Apply monitoring and logging techniques for deployed models 4. Develop strategies for model updates and versioning 5. Evaluate production models for performance and reliability	Docker FastAPI Flask MLflow BentoML AWS SageMaker

Concentration III: Business Intelligence and Advanced Data Analytics

NO	MATA KULIAH	LEARNING OUTCOMES	LEARNING TOOLS
1	Business Intelligence Tools 2 SKS	1. Implement data visualization and dashboard solutions using leading BI platforms 2. Apply data modeling techniques for business intelligence applications 3. Develop interactive reports and analytical applications for business users 4. Configure data connections and integration with various enterprise data sources 5. Evaluate BI tool capabilities for specific business requirements and user needs	Tableau Power BI Looker Metabase QlikView
2	Statistical Modeling 2 SKS	1. Apply appropriate statistical models to business data analysis problems 2. Implement regression, time series, and multivariate analysis techniques 3. Develop hypothesis testing frameworks for business decision support 4. Apply model validation and diagnostic techniques to ensure statistical validity 5. Evaluate statistical models for accuracy, robustness, and business applicability	R Python SPSS Stata Excel
3	Customer Analytics 3 SKS	1. Design customer segmentation models using demographic and behavioral data 2. Implement customer lifetime value analysis and churn prediction models 3. Develop marketing campaign effectiveness measurement frameworks 4. Apply customer journey mapping and touchpoint analysis techniques 5. Evaluate customer analytics initiatives for business impact and ROI	Python SQL Mixpanel Google Analytics HubSpot
4	Predictive Analytics 3 SKS	1. Design predictive modeling frameworks for business forecasting and planning 2. Implement machine learning algorithms for classification and regression problems 3. Develop feature engineering techniques to improve predictive model performance 4. Apply model ensemble methods to enhance prediction accuracy 5. Evaluate predictive models for accuracy, interpretability, and business utility	Python Scikit-learn XGBoost AutoML H2O.ai
5	Business Intelligence and Reporting 3 SKS	1. Design comprehensive BI architectures aligned with organizational strategy 2. Implement data governance frameworks for BI initiatives 3. Develop KPI frameworks and performance measurement systems 4. Create executive dashboards and automated reporting systems 5. Evaluate BI implementations for business impact, user adoption, and ROI	Power BI Tableau Looker SQL DAX
6	Supply Chain Management Systems 3 SKS	1. Apply analytics techniques to supply chain planning and optimization 2. Implement inventory management and demand forecasting models 3. Develop logistics network optimization and route planning solutions 4. Apply simulation techniques to evaluate supply chain performance 5. Evaluate supply chain analytics solutions for efficiency, resilience, and sustainability	SAP Oracle SCM Python Excel Arena Simulation

Mata Kuliah Wajib Semua Konsentrasi

NO	MATA KULIAH	LEARNING OUTCOMES	LEARNING TOOLS
1	Professional Ethics 2 SKS	1. Apply ethical frameworks to professional decision-making 2. Analyze ethical issues in technology and computing 3. Develop solutions to ethical dilemmas in professional contexts 4. Evaluate technology impacts on society and individuals 5. Demonstrate awareness of professional responsibility and accountability	Case Studies Notion IEEE/ACM Ethics Guidelines

SEMESTER 6 -- MBKM I 24 SKS • Data Science Project Management, ETL, Statistical Modeling & MK Umum

NO	MATA KULIAH	LEARNING OUTCOMES	LEARNING TOOLS
1	Data Science Project Management 4 SKS	1. Design comprehensive data science project plans including scope, resources, timelines, and risk assessments 2. Implement agile methodologies tailored for data science teams and workflows 3. Apply stakeholder management techniques for aligning data science initiatives with business goals 4. Develop data governance frameworks and documentation standards for data science projects 5. Evaluate data science project success using appropriate KPIs and metrics frameworks	Jira Notion Trello MS Project GitHub Projects
2	Advanced Data Processing and ETL 4 SKS	1. Design scalable ETL architectures for complex data integration scenarios 2. Implement advanced transformation logic for data normalization, cleansing, and enrichment 3. Apply performance optimization techniques for high-volume ETL processes 4. Develop monitoring and error handling frameworks for production ETL pipelines 5. Evaluate ETL solutions for reliability, maintainability, and processing efficiency	Apache Airflow Talend Python SQL dbt Spark

NO	MATA KULIAH	LEARNING OUTCOMES	LEARNING TOOLS
3	Data Wrangling and Preprocessing Techniques 4 SKS	1. Implement comprehensive data cleaning strategies for diverse data types (structured, semi-structured, unstructured) 2. Apply advanced techniques for handling missing data, outliers, and imbalanced datasets 3. Develop automated data quality assessment and validation frameworks 4. Implement feature engineering and transformation pipelines for machine learning preparation 5. Evaluate data preprocessing approaches for their impact on downstream analysis quality	Python Pandas NumPy OpenRefine Great Expectations
4	Statistical Modeling and Inference 4 SKS	1. Apply advanced statistical methods to complex real-world data problems 2. Implement various regression models, time series analysis, and multivariate techniques 3. Develop robust hypothesis testing frameworks with appropriate experimental designs 4. Apply Bayesian modeling approaches for uncertainty quantification 5. Evaluate statistical model assumptions, diagnostics, and performance metrics	Python R Stata SAS PyMC3 Statsmodels
5	Business Communication and Data Presentation 4 SKS	1. Design compelling data narratives that communicate insights to diverse stakeholders 2. Implement advanced data visualization techniques for complex data relationships 3. Develop executive dashboards and interactive reporting systems 4. Apply storytelling techniques to transform technical findings into business recommendations 5. Evaluate data presentations for clarity, impact, and actionability across different audience types	Power BI Tableau Google Slides Canva Prezi
6	Indonesian Way of Life / Pancasila 2 SKS	1. Explain the principles of Pancasila as Indonesia's state philosophy 2. Apply Pancasila values in professional and social contexts 3. Analyze social issues from a Pancasila perspective 4. Develop solutions to societal challenges based on Pancasila principles 5. Evaluate policies and practices in light of Pancasila values	Presentation Tools Discussion Forums
7	Religions of the World 2 SKS	1. Analyze the core beliefs and practices of major world religions 2. Compare and contrast religious perspectives on ethical issues 3. Apply religious and cultural understanding in diverse environments 4. Evaluate the influence of religion on society and individual behavior 5. Demonstrate respect for religious diversity and perspectives	Presentation Tools Research Databases

SEMESTER 7 -- MBKM II

24 SKS • ML Applications, Big Data, Predictive Analytics & Capstone

NO	MATA KULIAH	LEARNING OUTCOMES	LEARNING TOOLS
1	Machine Learning Applications 4 SKS	1. Design end-to-end machine learning solutions for complex business problems 2. Implement supervised, unsupervised, and reinforcement learning models for various domains 3. Develop feature engineering and selection strategies to optimize model performance 4. Apply model deployment and monitoring techniques in production environments 5. Evaluate machine learning solutions for accuracy, robustness, interpretability, and business impact	Python Scikit-learn TensorFlow PyTorch MLflow Docker
2	Big Data Technologies and Cloud Integration 4 SKS	1. Design scalable big data architectures using distributed computing frameworks 2. Implement data processing pipelines for batch and stream processing workloads 3. Apply cloud-native services for big data storage, processing, and analytics 4. Develop integration strategies between on-premises systems and cloud platforms 5. Evaluate big data solutions for performance, cost-efficiency, scalability, and security	Apache Spark Kafka AWS GCP Azure Databricks
3	Predictive Analytics and Forecasting 4 SKS	1. Apply advanced time series forecasting techniques for business planning and decision support 2. Implement ensemble methods and deep learning for complex prediction tasks 3. Develop predictive models for specialized domains (finance, marketing, operations) 4. Apply uncertainty quantification and scenario analysis in forecasting 5. Evaluate predictive models for accuracy, calibration, and practical utility in business contexts	Python Prophet ARIMA XGBoost Statsmodels
4	Data Ethics and Privacy 4 SKS	1. Apply ethical frameworks and principles to data science and AI initiatives 2. Implement privacy-enhancing technologies and anonymization techniques 3. Develop data governance policies that address ethical and privacy considerations 4. Apply regulatory compliance measures (GDPR, CCPA, etc.) in data projects 5. Evaluate data science solutions for fairness, transparency, accountability, and societal impact	OWASP ZAP Python Kali Linux GDPR Tools
5	Capstone Data Science Project 4 SKS	1. Design comprehensive data science projects addressing real-world problems 2. Implement end-to-end data pipelines from acquisition to insight generation 3. Apply appropriate analytical methods based on project requirements 4. Develop effective presentations and documentation of technical solutions 5. Evaluate project outcomes against business objectives and technical requirements	Full Stack Python Docker GitHub Cloud Platform
6	Applied Indonesian Language 2 SKS	1. Communicate effectively in professional contexts using Indonesian language 2. Apply appropriate Indonesian language styles for different business situations 3. Develop technical and business documentation in formal Indonesian 4. Analyze Indonesian language use in professional and academic contexts 5. Evaluate the quality and effectiveness of professional communication in Indonesian	MS Word LaTeX Turnitin
7	Civic / Kewarganegaraan 2 SKS	1. Analyze the relationship between citizenship rights and responsibilities in Indonesian society 2. Apply constitutional principles to contemporary civic issues and challenges 3. Evaluate the impact of public policies on citizens and communities 4. Develop strategies for civic engagement and participation in democratic processes 5. Demonstrate ethical decision-making aligned with national values and principles	Presentation Tools Research Databases

SEMESTER 8

10 SKS • Capstone Project, Thesis, Portfolio & Publikasi

NO	MATA KULIAH	LEARNING OUTCOMES	LEARNING TOOLS
1	Capstone Project 6 SKS	1. Design and execute a comprehensive project addressing a complex real-world problem in the chosen field of study 2. Apply appropriate methodologies, tools, and techniques from across the curriculum to develop innovative solutions 3. Evaluate the effectiveness of implemented solutions using appropriate metrics and assessment techniques 4. Demonstrate professional and ethical responsibility in all aspects of project development and execution 5. Synthesize knowledge from multiple disciplines to create an integrated approach to problem-solving	Full Stack Python Docker GitHub Cloud Platform CI/CD
2	Collaborative Project Komponen Capstone	1. Apply effective team collaboration skills in multidisciplinary project environments 2. Implement project management techniques for division of labor and task coordination 3. Resolve team conflicts constructively to maintain project progress 4. Demonstrate leadership and followership skills as appropriate in team contexts 5. Evaluate team processes and outcomes for continuous improvement	GitHub Jira Notion Slack MS Teams
3	Portfolio Komponen Capstone	1. Curate a professional portfolio that effectively showcases technical and creative capabilities 2. Document project work with clear explanation of problem statements, methodologies, and outcomes 3. Reflect critically on personal and professional growth throughout the educational program 4. Present work samples that demonstrate breadth and depth of professional skills 5. Design portfolio materials that communicate effectively to target audiences (employers, clients, etc.)	GitHub Pages Behance LinkedIn Personal Website
4	Product Prototype Komponen Capstone	1. Design and develop functional prototypes that demonstrate feasibility of proposed solutions 2. Apply iterative design and testing methodologies to refine prototype functionality 3. Implement appropriate technologies and techniques to create minimum viable products 4. Document technical specifications and development processes for knowledge transfer 5. Evaluate prototypes against user requirements and design specifications	Figma Docker GitHub Cloud Platform Postman
5	Scientific Publication Komponen Capstone	1. Design and conduct original research following scientific methodology 2. Analyze research data using appropriate quantitative and/or qualitative methods 3. Synthesize research findings in the context of existing literature and theoretical frameworks 4. Develop manuscripts that meet academic publication standards and conventions 5. Communicate research findings effectively to both specialist and non-specialist audiences	LaTeX Mendeley Zotero Google Scholar Turnitin

NO	MATA KULIAH	LEARNING OUTCOMES	LEARNING TOOLS
6	Thesis Komponen Capstone	1. Formulate research questions and hypotheses that address significant issues in the field 2. Apply appropriate research methodologies to investigate proposed questions 3. Analyze findings using critical thinking and appropriate analytical frameworks 4. Synthesize conclusions that contribute to the body of knowledge in the field 5. Defend research findings and conclusions through scholarly argumentation	LaTeXMendeleySPSS PythonTurnitin
7	Academic Writing in English 2 SKS	1. Apply academic writing conventions and formats 2. Develop well-structured arguments and thesis statements 3. Implement proper citation and referencing techniques 4. Analyze and synthesize information from academic sources 5. Evaluate and revise writing for clarity and coherence	LaTeXGrammarlyMendeley ZoteroMS Word
8	Research Methodology 2 SKS	1. Design research studies using appropriate methodologies 2. Apply data collection techniques relevant to research questions 3. Implement appropriate data analysis methods 4. Interpret research findings and draw valid conclusions 5. Communicate research results effectively through reports and presentations	SPSSPythonRLaTeX MendeleyGoogle Scholar

■ Ringkasan Total SKS

SEMESTER	SKS
Semester 1	20
Semester 2	20
Short Semester 1	9
Semester 3	18
Semester 4	18
Short Semester 2	9
Semester 5 (Concentration)	18
Semester 6 (MBKM I)	24
Semester 7 (MBKM II)	24
Semester 8	10
TOTAL	152 + 18 (Extra) = 170 SKS